

UST-MIL, Russian Federation

WMO: 310410

Lat: 59.639N

Lon: 133.111E

Elev: 588

StdP: 14.39

Time Zone: 9.00 (E09)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-59.1	-54.6	-61.8	0.1	-58.8	-57.2	0.2	-54.5	7.8	-24.4	6.9	-22.1	0.6	80	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	24.2	84.8	66.5	81.7	65.3	77.9	63.3	69.3	80.9	67.2	78.0	65.0	74.6	4.2	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.1	94.7	74.2	63.2	88.5	72.2	61.2	82.6	70.0	33.9	81.0	32.1	78.0	30.4	75.5	80.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.3	8.6	7.3	-61.2	90.8	6.0	3.9	-65.5	93.6	-68.9	95.9	-72.3	98.0	-76.6	100.9		
			-61.2	72.8	6.0	3.9	-65.5	75.6	-68.9	77.9	-72.3	80.0	-76.6	82.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	17.9	-34.6	-23.3	1.2	25.9	44.7	58.4	63.9	58.7	42.9	19.6	-14.5	-30.7	(6)
	DBStd	36.85	12.24	11.30	13.09	9.97	7.50	6.40	6.85	6.38	7.74	11.47	13.44	14.12	(7)
	HDD50	12736	2621	2051	1514	723	197	9	1	8	233	942	1934	2503	(8)
	HDD65	17291	3086	2471	1979	1173	630	213	105	212	663	1407	2384	2968	(9)
	CDD50	1026	0	0	0	0	32	262	433	279	20	0	0	0	(10)
	CDD65	106	0	0	0	0	0	16	72	18	0	0	0	0	(11)
	CDH74	1748	0	0	0	0	24	423	980	318	3	0	0	0	(12)
	CDH80	476	0	0	0	0	3	97	317	59	0	0	0	0	(13)
	WSAvg	2.8	1.4	1.8	2.6	3.6	4.1	3.6	3.1	3.0	3.1	2.9	2.2	1.9	(14)
	PrecAvg	18.9	0.7	0.5	0.6	0.9	1.9	2.0	3.0	2.7	2.5	2.0	1.3	0.8	(15)
(16) Precipitation	PrecMax	27.8	1.4	1.1	2.5	2.9	2.9	4.1	6.6	6.8	5.0	5.5	3.0	1.7	(16)
	PrecMin	13.4	0.3	0.2	0.1	0.2	0.5	0.8	1.0	0.8	0.7	0.7	0.3	0.3	(17)
	PrecStd	3.3	0.3	0.2	0.5	0.5	0.7	0.7	1.1	1.3	0.9	0.9	0.7	0.4	(18)
	PrecStd	3.3	0.3	0.2	0.5	0.5	0.7	0.7	1.1	1.3	0.9	0.9	0.7	0.4	(18)
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.8	12.2	41.9	55.4	77.1	86.3	91.8	84.8	71.3	48.7	31.7	14.4	(19)
		MCWB	1.3	10.8	34.5	43.3	58.0	65.9	69.8	67.4	58.1	40.8	31.2	14.1	(20)
	2%	DB	-3.9	4.9	35.2	50.4	70.3	82.8	86.8	81.5	65.8	43.0	19.6	1.5	(21)
		MCWB	-4.2	3.8	29.9	39.8	53.7	63.3	67.3	66.1	54.3	38.2	17.9	0.9	(22)
	5%	DB	-11.2	0.1	30.0	46.9	66.0	78.9	83.4	77.8	61.1	38.5	11.0	-3.7	(23)
		MCWB	-11.7	-0.7	25.4	37.7	51.0	61.2	66.8	64.5	52.4	35.0	10.0	-4.1	(24)
	10%	DB	-16.4	-5.4	25.3	43.1	61.5	75.3	79.4	73.4	56.8	34.9	4.7	-9.6	(25)
		MCWB	-16.7	-6.0	21.8	35.1	48.8	59.8	65.3	61.5	49.3	32.3	3.9	-10.1	(26)
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.5	10.8	35.0	43.9	58.9	68.9	72.4	70.6	58.9	42.5	31.4	14.2	(27)
		MCDB	1.8	12.0	41.6	54.4	77.1	80.8	86.2	80.7	67.7	46.4	31.8	14.4	(28)
	2%	WB	-4.2	4.1	29.9	40.8	55.2	65.9	70.2	67.8	56.1	38.9	18.0	1.1	(29)
		MCDB	-3.9	5.0	34.7	49.5	68.6	78.9	82.8	79.2	63.5	42.5	19.3	1.7	(30)
	5%	WB	-11.5	-0.7	25.7	38.2	52.1	63.1	68.5	65.5	53.5	35.5	10.1	-4.0	(31)
		MCDB	-11.2	0.1	29.8	46.2	62.8	75.1	79.6	74.8	59.4	38.0	11.0	-3.7	(32)
	10%	WB	-16.7	-6.1	21.9	35.3	49.7	61.3	66.5	63.0	50.9	32.5	4.0	-10.0	(33)
		MCDB	-16.4	-5.5	25.3	42.5	60.2	72.4	76.8	70.9	55.7	34.9	4.7	-9.6	(34)
(36) Mean Daily Temperature Range	5% DB	MDBR	14.6	24.1	32.2	25.9	23.3	26.6	24.2	24.0	21.5	16.5	16.9	13.5	(35)
		MCDBR	19.9	26.3	32.5	27.4	33.7	35.4	31.3	31.0	29.5	18.1	21.5	18.1	(36)
		MCWBR	19.5	25.7	27.1	18.6	19.6	17.8	15.0	16.9	18.7	13.4	20.6	17.7	(37)
	5% WB	MCDBR	20.0	26.7	30.6	25.9	29.6	29.0	26.6	26.1	25.1	15.7	21.6	17.7	(38)
		MCWBR	19.6	25.7	25.4	17.8	17.8	16.3	13.9	15.2	16.6	11.8	20.8	17.6	(39)
(40) Clear-Sky Solar Irradiance	taub		0.235	0.271	0.313	0.389	0.455	0.439	0.521	0.458	0.353	0.307	0.245	0.211	(40)
	taud		2.005	2.055	2.033	1.955	2.031	2.192	1.952	2.109	2.390	2.272	2.032	1.938	(41)
	Ebn at Noon		170	229	254	252	244	252	224	230	245	216	161	131	(42)
	Edh at Noon		16	26	36	48	48	42	52	41	26	21	15	11	(43)
(44) All-Sky Solar Radiation	RadAvg		181	523	1074	1599	1644	1872	1712	1330	847	395	229	102	(44)
	RadStd		9	22	40	70	134	124	101	109	52	53	18	4	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (4 neighbors)	+0.92	N/A	N/A	N/A	N/A	N/A	N/A	-295	N/A	N/A

Nomenclature: See separate page